

JOENSUU AP, Finland

WMO: 029290

Lat: 62.660N

Lon: 29.611E

Elev: 398

StdP: 14.49

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.7	-11.4	-22.7	1.6	-17.3	-16.1	2.3	-11.4	17.9	21.8	16.3	22.5	2.8	300	0.636

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	80.4	65.1	76.8	63.2	73.4	61.7	67.5	75.8	65.5	73.5	63.6	70.9	8.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	92.2	70.8	62.5	85.8	69.2	60.5	80.0	67.3	32.2	75.3	30.5	73.3	29.1	70.8	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.5	14.8	13.6	DB	-21.4	84.6	7.5	4.7	-26.8	88.0	-31.2	90.7	-35.5	93.3	-41.0	96.7	(4)
(5)			WB	-21.7	69.9	7.5	2.8	-27.1	71.9	-31.4	73.5	-35.6	75.1	-41.0	77.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.3	15.5	15.8	24.3	36.2	47.9	56.9	63.0	59.0	49.8	38.6	28.9	21.8		
	DBStd	18.64	13.73	13.07	8.81	6.96	8.52	6.86	6.06	5.74	6.06	7.12	8.54	12.10		
	HDD50	5347	1069	959	797	418	142	15	1	4	77	357	632	876		
	HDD65	9838	1534	1379	1262	865	533	257	110	200	456	819	1082	1341		
	CDD50	1065	0	0	0	3	76	223	404	284	72	3	0	0		
	CDD65	79	0	0	0	0	2	14	48	15	0	0	0	0		
	CDH74	612	0	0	0	0	37	110	361	104	0	0	0	0		
	CDH80	99	0	0	0	0	2	15	69	13	0	0	0	0		
	WSAvg	6.7	6.7	6.6	6.8	6.3	6.9	6.8	6.3	6.0	6.5	7.0	7.3	7.4		
	PrecAvg	24.7	1.6	1.3	1.3	1.3	1.5	2.4	2.9	3.3	2.4	2.4	2.2	1.9		
	PrecMax	33.7	3.5	3.0	3.4	3.1	3.4	5.2	6.9	8.1	5.7	5.0	5.7	3.5		
	PrecMin	15.7	0.3	0.2	0.2	0.2	0.1	0.1	0.6	1.0	0.7	0.4	0.5	0.6		
	PrecStd	4.4	0.8	0.7	0.6	0.7	0.7	1.2	1.5	1.6	1.0	1.1	0.9	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.4	39.0	46.7	63.4	78.8	82.7	85.5	81.4	69.5	55.4	47.4	42.8		
		MCWB	36.2	36.2	39.2	51.1	58.2	63.2	68.5	66.2	60.2	52.2	46.1	41.9		
	2%	DB	35.3	35.5	41.3	56.3	72.4	77.3	81.6	76.8	64.7	52.3	44.4	39.1		
		MCWB	34.5	34.5	36.0	46.7	56.8	62.3	66.3	63.7	58.0	50.5	43.2	38.4		
	5%	DB	33.6	33.6	37.7	51.8	67.0	73.3	78.2	73.2	62.0	50.7	42.0	36.7		
		MCWB	33.2	32.9	34.0	43.3	53.5	60.4	64.9	62.3	56.6	49.0	41.2	35.8		
	10%	DB	32.0	32.1	35.4	47.7	62.8	69.7	74.8	69.6	59.3	48.6	39.6	34.7		
		MCWB	31.7	31.7	32.8	40.4	51.2	58.2	63.2	60.7	55.1	47.2	38.8	34.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.4	36.6	39.8	52.8	61.9	67.8	71.0	68.9	61.4	53.5	46.3	41.7		
		MCDB	36.8	37.9	45.4	60.2	74.0	76.2	80.5	78.1	65.9	54.9	47.1	42.3		
	2%	WB	34.4	34.4	36.8	48.2	58.4	65.2	68.6	65.8	59.5	51.3	43.2	38.3		
		MCDB	34.8	35.1	40.3	54.7	68.0	73.5	77.5	72.9	63.3	52.3	43.9	38.7		
	5%	WB	33.0	33.0	34.8	44.0	55.5	62.4	66.6	63.9	57.5	49.4	41.1	35.8		
		MCDB	33.3	33.4	36.9	50.2	64.9	70.8	75.3	70.8	60.8	50.5	41.7	36.3		
	10%	WB	31.8	31.7	33.3	41.0	52.5	59.5	64.7	61.9	55.7	47.4	38.9	34.0		
		MCDB	32.1	32.2	34.8	46.2	61.0	67.7	72.6	68.0	58.7	48.4	39.6	34.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	11.2	13.8	14.9	17.4	16.0	15.8	14.7	12.2	8.3	6.9	8.9		
		MCDBR	7.2	7.5	15.4	21.2	24.7	21.0	20.7	19.8	15.7	9.3	7.1	6.9		
		MCWBR	7.1	7.2	11.1	12.8	12.0	9.5	8.6	9.3	9.8	7.7	7.0	6.7		
	5% WB	MCDBR	7.1	7.7	12.5	20.0	20.9	18.3	18.3	17.3	13.7	8.6	6.9	6.9		
		MCWBR	7.1	7.5	9.6	12.5	11.4	9.3	8.6	9.0	9.5	7.7	6.9	6.8		
(40) Clear-Sky Solar Irradiance	taub		0.231	0.285	0.312	0.351	0.350	0.345	0.370	0.353	0.332	0.312	0.248	0.211		
	taud		2.007	2.112	2.184	2.320	2.453	2.495	2.437	2.468	2.516	2.402	2.098	1.971		
	Ebn at Noon		129	205	248	264	276	279	267	262	247	202	122	73		
	Edh at Noon		11	22	30	32	31	30	31	28	22	17	10	6		
(44) All-Sky Solar Radiation	RadAvg		64	243	724	1163	1528	1683	1601	1221	677	259	66	29		
	RadStd		8	58	96	140	174	95	153	141	79	41	11	4		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (28 neighbors)	+0.97	N/A	N/A	N/A	N/A	N/A	-344	N/A	N/A	N/A

Nomenclature: See separate page